Exp Fluids
DOI 10.1007/s00348-011-1085-3

RESEARCH ARTICLE

Residence time measurement of an isothermal combustor flow field

Liangta Cheng · Adrian Spencer

Received: 30 September 2010 / Revised: 8 March 2011 / Accepted: 20 March 2011
© Springer-Verlag 2011

Abstract Residence times of combustors have commonly been used to help understand NO_x emissions and flame blowout. Both the time mean velocity and turbulence fields are important to the residence time, but determining the residence time via analysis of a measured velocity field is difficult due to the inherent unsteadiness and the three-dimensional nature of a high-Re swirling flow. A more direct approach to measure residence time is reported here that examines the dynamic response of fuel concentration to a sudden cutoff in the fuel injection. Residence time measurement was mainly taken using a time-resolved planar laser-induced fluorescence (PLIF) technique, but a second camera for particle image velocimetry (PIV) was added to check that the step change does not alter the velocity field and the spectral content of the coherent structures. Characteristic timescales evaluated from the measurements are referred to as convection and half-life times: The former describes the time delay from a fuel injector exit reference point to a downstream point of interest, and the latter describes the rate of decay once the effect of the reduced scalar concentration at the injection source has been transported to the point of interest. Residence time is often defined as the time taken for a conserved scalar to reduce to half its initial value after injection is stopped: this equivalent to the sum of the convection time and the half-life values. The technique was applied to a high-swirl fuel injector typical of that found in combustor applications. Two test cases have been studied: with central jet (with-jet) and without central jet (no-jet). It was found that the relatively unstable central recirculation zone of the no-jet case resulted in increased transport of fuel into the central region that is dominated by a precessing vortex core, where long half-life times are also found. Based on this, it was inferred that the no-jet case may be more prone to NO_x production. The technique is described here for a single-phase isothermal flow field, but with consideration, it could be extended to studying reacting flows to provide more insight into important mixing phenomena and relevant timescales.

Extended version of Lisbon 2010 Paper #1736 for consideration of Special Issue in Lisbon 2012, Revision 2, March 2011.

Electronic supplementary material The online version of this article (doi:10.1007/s00348-011-1085-3) contains supplementary material, which is available to authorized users.

L. Cheng · A. Spencer (✉)
Department of Aero & Auto Engineering,
Loughborough University, Loughborough, UK
e-mail: a.spencer@lboro.ac.uk

1 Introduction

Measurement of residence time and temperature is of practical interest in gas turbine combustor research because the two parameters have usually been used to correlate with various emission indices. For example, Roffe and Venkataramani (1978) showed for a propane flame that the NO emission index, NO_{EI}, was described well by the empirical expression:

$$\ln(NO_{EI}/t_G) = -72.28 + 2.80\sqrt{T_{AF}} - T_{AF}/38 \quad (1)$$

where T_{AF} is the adiabatic flame temperature (Kelvin), t_G is the characteristic combustor residence time (ms), and numerical constants have units consistent with the equation. The residence time only becomes insignificant for very lean mixtures (equivalence ratios below ~ 0.4) because the temperature, which determines the equilibrium point of the dissociation reaction (e.g., NO production from

Published online: 12 April 2011

O_2 and N_2 in the Zeldovich chain mechanism (Heywood et al. 1970)), is then very low. In a relevant study, Driscoll et al. (1992) showed that the Damkhöler number, which may be interpreted as the ratio of residence time, τ_{res}, and chemical kinetic time, τ_{chem}, correlated well with NO_x emission data in hydrogen and helium flames that have simple chemistry. The Damkhöler number, as defined in Eq. 2, has also been used to predict flame blowoff with success (Lieuwen 2006; Noble et al. 2006):

$$Da = \frac{\tau_{res}}{\tau_{chem}} = \frac{S_L^2 d}{\alpha U_{ref}} \tag{2}$$

where S_L is the laminar flame speed, α is the thermal diffusivity, U_{ref} is the reference velocity (e.g., temperature-scaled fuel velocity), and d is typically taken to be the length of the combustor or recirculation zone. Hence, residence time under this definition ($\tau_{res} = d/U_{ref}$) simply refers to the convective time it takes to complete one through flow or recirculation in the case of a flame holder. Although such global measures of residence time allow correlations of emissions to be made and hence allow combustor scaling to be done, they do not help identify the regions of emission production. Such estimates of τ_{res} are also questionable because there are many routes via which fluid could travel through the combustor, and the length of recirculation zone could also be subject to inconsistencies in interpretation. Other methods to estimate residence times in a combustor have been reported. For example, residence time can be inferred from tracing particles through a measured (Giordano et al. 2001) or CFD-predicted velocity fields (Di Nardo et al. 2009). Ozdemir and Peters (2001) used T_iO_2 as a particulate passive tracer in the mild combustion reaction zone of a combustor. Concentration was measured at a point via Mie-scattering in response to ceasing the tracer injection and referred to as a "step-down" procedure. Highly skewed residence time distributions were observed with the PDF biased strongly toward long residence times. First- through fourth-order statistics of residence time distributions were calculated and discussed in the context of NO_x emissions. The long time average of the residence time was shown to be a misleading quantity when strongly skewed distributions were observed. This is particularly true if the tail of the PDF has timescales associated with it that are significantly longer than the chemical kinetic timescales when the average or global residence time is not. However, these methods can be questioned due to the differences between the equations governing the transport of momentum and of a conserved scalar: The former is influenced by the pressure field, while the latter is not; turbulent diffusion rates of the two quantities will therefore be different (Pantano et al. 2003). The estimation of residence time from conserved scalar tracking, on the other hand, has also been reported. For example, Arai et al. (1998), followed by similar work of van der Lans et al. (1997), used helium as a tracer and injected it into a swirl-stabilized combustion chamber with and without reaction. The helium injection was stopped suddenly, and the attenuation curves (helium concentration decay) at various locations were studied in response. The authors observed slightly shorter residence times due to heat release as would be expected due to flow acceleration caused by density reduction. Accuracy of the absolute concentration measurement was relatively poor in absolute terms ($\sim$20%), but it is the rate of concentration change, which was found to be of importance to the technique. Despite these findings, the two aforementioned techniques were single-point-based measurements that do not help identify the source of emissions without extensive measurement traversing campaigns.

A simple and efficient technique is reported here to provide spatially resolved estimates of residence time using temporally resolved planar-induced fluorescence (PLIF). This measurement approach, applied to a representative high-swirl combustor flow, examines the dynamic response of non-reacting and isothermal fuel concentration after the injection of fuel is suddenly switched off. It is believed that this is the first time that planar LIF has been used in this step-down approach, particularly for a combustion-related flow. Previous work that has been reported has utilized point measurements, non-simultaneously and at low spatial resolution.

2 Experimental setup

Measurements were taken in water under isothermal conditions. Using water as the working fluid has many advantages. First, seeding particles are prone to slippage in air while they are neutrally buoyant in water. Second, water allows gravity to be utilized as the stable driving force, thereby eliminating fluctuations in the pump delivery rate by the use of a constant head tank with an overflow branch. Finally, if geometrical scale (d) and Reynolds number ($U_{ref}d/\nu$) are matched (along with careful considerations of the momentum flux ratio between the swirl stream and the central jet stream), then the water test reference velocity will be in the opposite ratio to the kinematic viscosity (ν) of air and water, which is 14.6 at standard laboratory conditions. Thus, for a flow feature of a particular Strouhal number ($f \cdot d/U_{ref}$), the actual frequency in water will be lower than that in air by this same factor. This is beneficial to measurement devices of low temporal resolution relative to typical timescales in air flows: The 500 Hz data capture rate used here is approximately equivalent to 7.3 kHz in air flow at the same Reynolds number and geometrical scale.

2.1 Test rig

The test rig is an isothermal, vertically flowing, closed-loop, 5-m-tall constant head water rig capable of delivering up to 15 kg/s to the test section. A full description of the facility can be found in Spencer (1998) or more briefly in Spencer et al. (2008). For concentration measurements, a dye injection system is required to simulate the fuel feed. Details of the dye system and its performance have been documented in Midgley (2005). Briefly, the system consists of a header tank that contains up to 30 l of dye solution of a fixed, well-mixed Rhodamine B concentration. A pump is used to overcome pressure loss in the pipe work and to deliver the solution to the test section. Dyed water mass flow rate is monitored via a flow meter calibrated against PIV and direct collection of the delivered flow. An electrical switch allows instantaneous opening and closing of the solenoid valve that controls the flow in the fuel feed hose pipe.

2.2 Fuel injector test cases

The test case is a partially premixed fuel injector for the combustor of an industrial gas turbine based on the Turbomeca lean premix design, as shown in Fig. 1a. The flow streams indicated show fuel injected into the annular swirl stream with a central air jet that can be stopped to produce the no-jet case. The fuel and air streams are found to be not fully premixed at the annulus exit, exhibiting some radially biased unmixedness. The injector geometry is modified only slightly to accommodate high optical access for the PIV/LIF measurements and also provides a well-defined exit flow separation point, as shown in Fig. 1b. The original measurements were motivated by a need to provide validation data for Large Eddy Simulation studies; thus, this geometrical change also eased the computational expense of modeling the exit flow separation point using LES. The fuel injector has been tested previously in the non-premixed mode where fuel is introduced via the central jet only. Here, a simple modification was made to inject the dye solution into the premixing cavity via a single pipe. After filling up the cavity, the dye exits through the 12 premixing ports into the radial entry slots, such that some premixing takes place with an air-to-fuel mass flow ratio of 20.1–1.

Two test configurations have been studied: with central jet (with-jet) and without central jet (no-jet). The mass flow split (central jet to swirl stream) in the with-jet case was determined by matching the momentum flux ratio of a combusting flow as reported in Janus et al. (2007). Meier et al. (2007) have also investigated the reacting no-jet case. The radially fed swirler has an annular swirl stream outer diameter $D_s = 37.63$ mm at the fuel injector exit, which for convenience was at 1.35:1 scale compared to the experiments of Janus and Meier. The inner diameter of the annulus was formed by the outer wall of a central jet nozzle inner diameter $D_j = 5.4$ mm. The test conditions corresponded to swirler and jet exit Reynolds numbers of 7.48×10^4 and 2.63×10^4, respectively. *Re* is here defined using D_s and D_j and the bulk mean axial velocities based on geometrical exit areas: $V_{x,j} = 4.88$ m/s, $V_{x,s} = 1.99$ m/s. Note, however, that the discharge coefficient of the swirl passage was measured to be 0.5, so the peak velocity in the swirl stream is about $1.5V_{x,s}$. It can be seen in Table 1 that because the axial momentum flux ratio ($M_{x,j}/M_{x,s}$) has been matched, the velocity ratio and mass flow rate ratios (m_j/m_s) for the with-jet condition are slightly different to the engine condition because the engine condition has a density difference between the jet and swirl streams not present in the water flow rig. This also means that there is some *Re* number discrepancy, but additional tests at ¾ and ½ *Re* showed no discernable

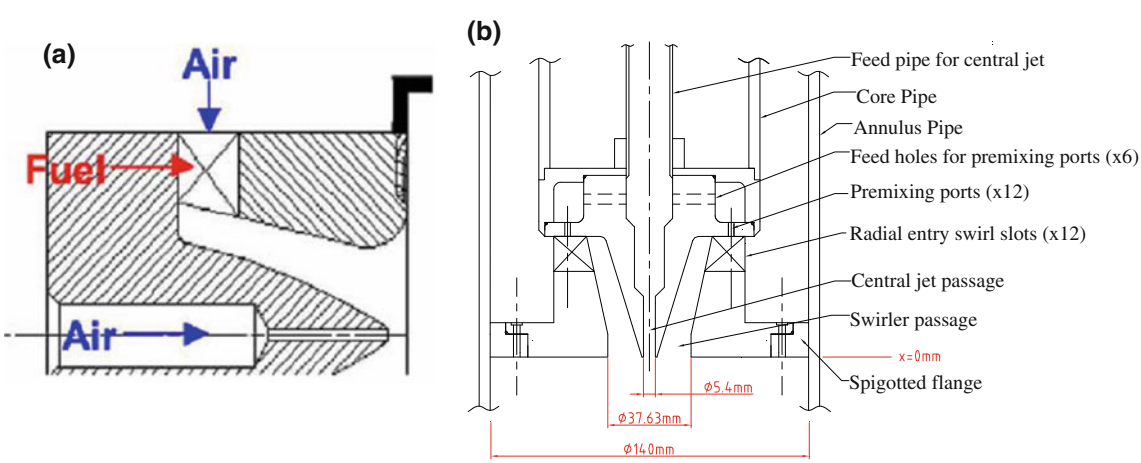


Fig. 1 Fuel injector geometry: **a** partial premixed engine configuration **b** test section

Table 1 Comparison of isothermal experimental test conditions with reacting cases

Condition	$V_{x,s}$ (m/s)	$V_{x,j}/V_{x,s}$	$Re_s \times 10^4$	Re_j	m_j/m_s	$M_{x,j}/M_{x,s}$
Janus et al. (2007)	37.9	2.24	2.9	3.3×10^4	0.053	0.124
Meier et al. (2007)	~24	0	3.5	0	0	0
With-Jet	1.99	2.41	7.48	2.63×10^4	0.0514	0.124
No-Jet	1.99	0	7.48	0	0	0

effect on the resultant velocity field. A higher *Re* was chosen, and desirable, to ensure that the rig inlet flow conditions were suitably away from transitional effects to aid with inlet condition definition for subsequent CFD predictions. See Midgley (2005) for a detailed description of these scaling issues. No correction is made for the fuel/dye shutoff which could reduce the swirl stream mass flow by 5%: Once the jets in cross-flow are removed from the swirl slots, their blockage effect is also removed, and the PIV measurements indicate a smaller change in mean velocity of less than 3%. The fuel/dye injection momentum ratio is approximately 5:1 based on bulk average velocities, though the velocity distribution in the swirl slots is unknown. The level of swirl that a flow contains can be assessed via the Swirl number, which is defined as the axial flux of swirl momentum divided by the axial flux of axial momentum times a characteristic radius, ($D_s/2$), Chigier and Beer (1972). Extensive PIV measurements have shown that the resultant swirl number for this injector is 0.74.

2.3 Optical arrangement and instrumentation

Combined 2C-PIV/PLIF was carried out in the x-r plane with two cameras placed perpendicular to each other, and their distances from the dichroic image splitter were adjusted so that they both viewed the same area on the calibration plate, as shown in Fig. 2. The fields of views (FoVs) of the two HighSpeedStar 6 cameras were mapped

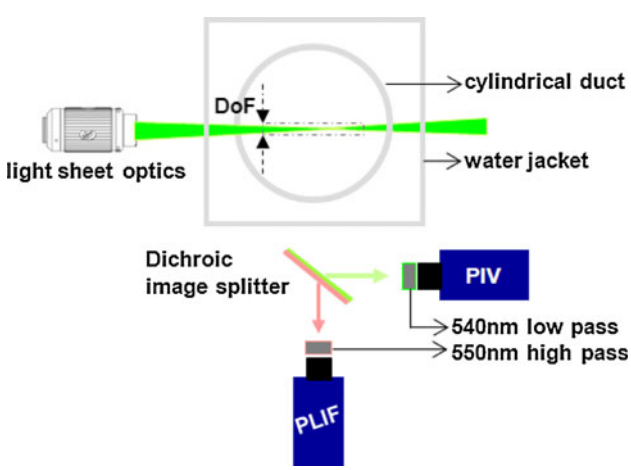


Fig. 2 Combined 2-component-PIV/PLIF optical arrangement

using the calibration algorithm provided by LaVision DaVis 7.2. Thus, the FoV of around 60 × 60 mm was resolved by 1,024 × 1,024 pixels at 10-bit resolution, giving 1024 grayscales. The area of interest was illuminated by a pulsed laser, generated by the Quantronix Darwin Duo Nd-YLF 30 mJ laser system. The Depth of Focus (DoF) was maintained to be just larger than the laser sheet thickness of 1 mm. Thus, the minimum measurement volume for scalar measurements was approximately $0.06\text{mm}^2 \times 1$ mm. The PIV was carried out simultaneously with PLIF to check that the sudden fuel cutoff does not have a pronounced effect on the velocity time history. Polyamide spheres of 20 μm in diameter were used as seeding particles.

2.4 PIV and PLIF processing

To minimize errors in the PIV-deduced velocity, parameters such as particle image diameter and inter-frame time were optimized using an established guideline (Adrian 1991) for error reduction. A multipass (64 × 64 pixels interrogation cell size in the first pass and 32 × 32 in the second) was applied with a 50% overlap along with other optimization parameters in the DaVis 7.2 software. The velocity fields presented here and the PIV methods used to obtain them have previously been well documented, see Midgley (2005) for further details.

Rhodamine B ($C_{27}H_{31}ClN_2O_3$), abbreviated as Rh-B, is well suited for concentration measurements due to its high solubility in water and its fluorescence stability. Several steps were performed on the raw intensity field. First, the background intensity, time averaged over 20 images, was subtracted to eliminate the camera's dark image intensity, background noise, and non-fluorescent light that leaked through the filter lens. To correct for spatial variation of light sheet energy (due to Gaussian beam intensity and sheet spreading), DaVis' sheet correction algorithm was applied. The Rh-B concentration used was low enough (~20 μg/l) such that the intensity decay in the sheet propagation direction was within 8% with the test section completely filled with the reference dye solution (the concentration in the final test is expected to be much lower due to mixing with water in the swirl stream). Hence, no absorption correction was required or applied. Finally, a

3 × 3 pixel averaging with uniform weighting was used as a spatial filter to reduce contamination by light sheet streaks (caused by variations in Perspex wall optical quality) and pixel sensitivity noise. This results in a 0.18 mm^2 × 1 mm measurement volume. The concentration calibration consisted of measuring intensity in the central region of the FoV when the test section was filled with dyed fluid at 10 concentration levels. A linear relationship between the resulting 10 intensity levels (photon counts) and concentration was obtained. Fluctuation in the light sheet energy was measured by recording 100 images and calculating its rms, which was found to be around 2% of the mean.

3 Measurement technique

For a near one-dimensional flow with gradual area increase, it may be expected that the arrival time at point B of the impulse of a tracer released at point A would have a near Gaussian distribution, as shown on the right of Fig. 3, with mixing happening equally in the upstream and downstream directions. If the flow deceleration is instead performed suddenly, then recirculations will be formed as in Fig. 4. Now, the impulse released at A may become entrained into the recirculations, and the transport time to the exit plane of some of the tracer will be considerably increased, particularly at C where the streamlines interact strongly within the recirculation shear layer. This will skew the arrival time distribution at the exit plane as shown. If a steady input followed by a step reduction of trace gas is used at A rather than an impulse, then the response at the exit plane will be (one minus) the cumulative arrival time distribution derived from the impulse. Thus, at B and C, the tracer concentration will remain high until the step has been transported to that location. Because of mixing, however, there will be a characteristic decay curve dependent upon how the history of the flow has affected its transit time. By calculating the normalized gradient of the cumulative arrival distribution in Fig. 4c, the arrival or residence time distribution (RTD) can be derived corresponding to the curve in Fig. 4b, see van der Lans et al. (1997) for example.

For typical combustor swirling flows, it would be expected that recirculations feature strongly. Here, scalar responses were found to be well approximated by an exponential decay for most of the measurement points: supported by the findings of Ozdemir and Peters (2001) and others as discussed in the introduction. Ozdemir and Peters (2001) described his point measurements of residence times in terms of nth order statistics. The first being equivalent to a mean convection time, the second a rms of the residence time distribution, and the third being the skewness of this distribution. The first- and second-order statistics were found to be of similar importance to understanding NOx formation in the mild combustor. The skewness indicated that the residence time distributions were far from Gaussian; thus, the (second order) rms value had a vague relevance to the distribution curve. van der Lans et al. (1997) employed a similar point-based step-down procedure using a helium tracer applied to a geometry not dissimilar to the one employed here involving swirl-induced vortex breakdown and associated recirculation zones. Their helium concentration decay curves were similar to curve C of Fig. 3c, and they attempted to determine the residence time distribution curve with some success by differentiating the normalized (by the initial value) decay curve, which appeared close to exponential in nature. Because of flow turbulence, the repeatability was poor when differentiating the decay signal that contains steep gradients so some temporal filtering was required. Stability behind flame holders has received attention in the

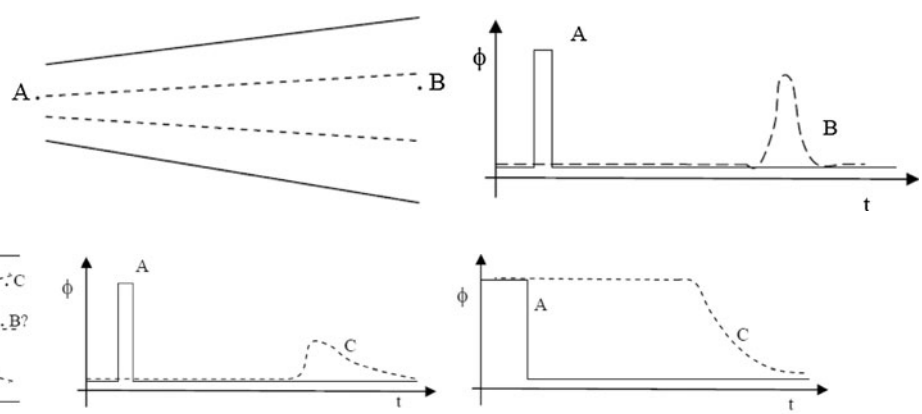


Fig. 3 Flow without recirculation; scalar response at B to impulse at A

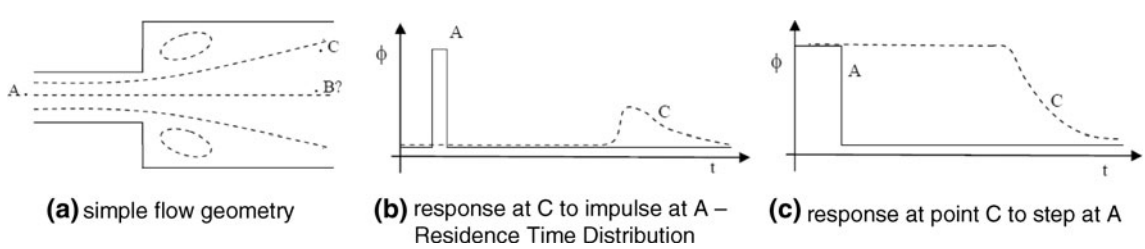


Fig. 4 Two-dimensional flow with recirculation; conserved scalar response

past by Winterfield (1965), among others, who have also shown that exponential decay is a reasonable model for the rate at which a tracer concentration reduces from a recirculation trapped behind a triangular flame holder. To avoid the problems of repeatability, it is thus attempted here to parameterize the decay curve as a "convection time" over which the scalar concentration remains statistically constant after the tracer injection has been stopped, followed by a "decay time" or "half-life" as determined from an exponential fit to the decaying curve after the convection time period has ended. When measuring a combustor global residence time using a probe at the combustor exit plane, the usual definition is the time delay from stopping the injection of the tracer at inlet to the point at which the concentration at the exit probe has reached half its initial value. It can be seen that this will correspond to the sum of the convection time and the half-life for the parameterized curve proposed here. There should be good, near linear, agreement between the measured value and a "global" combustor residence time based on suitable reference velocity and length scale that is often used to correlate emissions performance as in Eq. 1.

Thus, the technique used here is essentially represented by Fig. 4c. The rig is set running with the required test conditions for a few seconds until steady condition is reached, after which the recording sequence was started with the pulsed laser and cameras operating at 500 Hz (allowing a total recording time of 6.144 s for 3,072 image pairs). Then, the electrical switch connected to the solenoid valve described is used to shut the fuel feed. A switch-triggered laser pointer then turns on and projects a beam into the field of view, causing a slight brightening in a spot near the edge of the FOV, marking the time instant at which the dye was shut. After the dye shut signal is registered, there is a characteristic response time of the injection and pre-mixing system. This depends on the valve response time and flow paths from the fuel injection point to the exit of the swirler. These aspects are not investigated here, and this characteristic delay time is subtracted from the results presented. In this way, the residence times will refer to the period from which the air is directly exposed to high temperatures, with the time origin starting as the fluid exits the premixing duct.

3.1 Characteristic times

After the response period, the concentration field starts to decay very rapidly near the fuel injector exit until a near-zero concentration is reached throughout majority of the FoV. The time at which the response period ends (i.e., the starting point of concentration decay at the swirl stream exit) is taken to be the reference value. The difference between the time for concentration to begin falling at

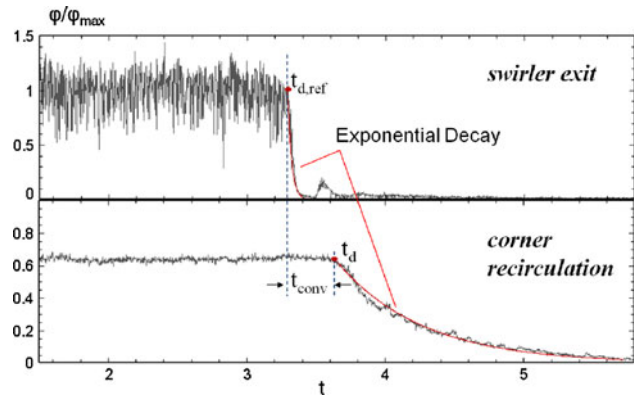


Fig. 5 Illustration of characteristic times

another location and that at the reference point is then interpreted as the convection time, as shown in Fig. 5. The reference starting point of decay, $t_{d,ref}$, is determined by examining the time histories of scalar concentration at the exit. For other downstream points in the FoV (e.g., the corner recirculation also shown in Fig. 5), t_d is detected using an algorithm. It first identifies when the concentration last drops–and remains–below one standard deviation of the mean value after $t_{d,ref}$, and then, an exponential fit is applied to the data after this point. The fit coefficients of this exponential curve provide the location in time at which the model curve intercepts the steady time average intensity. The curve originating from this crossing point is then fitted again to the concentration decay signal, producing a second set of curve fit coefficients. This step is iterated until the location of the intercept point of the time mean and exponential curve converges, typically in less than 10 iterations.

As shown in Fig. 5, the rates of decay for the two points are also significantly different. An exponential function has therefore been fitted to describe the rate of decay for all locations:

$$I(t) = I_0 e^{-b(t-t_d)} \tag{3}$$

where I_0 is the initial time mean intensity (concentration) before decay occurs, the time t_d coincides with the intercept point found, and b is the decay coefficient. The exponential decay coefficient, b, is converted to a half-life time using $t_{1/2} = \ln(2)/b$, which gives the time interval required for the concentration, once started, to decay to half of its original value.

The two timescales characterized here, namely convection time and half-life time, can then be normalized to give a dimensionless values using $\tau = V_{x,s}.t/D_s$, which allows the results to be scaled from the experiment onto various combustor conditions using a reference velocity and dimension. For the present work, V_{xs} and D_s are the swirl stream axial velocity and swirler diameter, respectively, as given in the test conditions. Thus, the non-dimensionalized times

presented are the convection time τ_{conv}, the half-life time $\tau_{1/2}$, and the sum of these two, the residence time, τ_{res}. This terminology is used because this will correspond to the maximum point on the equivalent residence time distribution. Referring back to Fig. 4(b), this is the time difference in time between the peaks of the two scalar concentration curves at A and C.

4 Results

A detailed description of the mean velocity and turbulence intensity fields of the with-jet and the no-jet cases can be found in Midgley and Spencer (2005), Spencer et al. (2008), Dunham et al. (2009) and of the steady-state scalar mixing in Cheng et al. (2010). Key findings from this previous work will be mentioned briefly here, but it is worth noting that the velocity field results from the non-reacting isothermal water tests of this work were in very close agreement with those of the combusting air test (Janus et al. 2007) in terms of both time-averaged and unsteady flow features, including the spectral content. Two instantaneous PIV velocity fields for the with-jet case are shown in Fig. 6 taken from Midgley and Spencer (2005). These are from two perpendicular planes: a plane containing the axial centerline and a radial axis and the other is coincident with the injector exit plane containing the radial and circumferential directions. These authors have identified two vortex pairs that originate from inside the swirler passage, the stronger pair of which can be clearly seen in cross-section in Fig. 6b. These follow helical paths as they travel downstream, entrained in the inner shear layer of the conical jet issuing from the swirl stream, evident as periodic structures in Fig. 6a. This vortex pair is best visualized in 3-dimensions, shown here in Fig. 6c as an isosurface of correlation in the radial velocity taken from LES predictions of Dunham et al. (2009). The extent over which velocities correlate to a reference point in the injector exit signifies the level of coherence and periodicity present. A much weaker pair of vortices also exists 90° out of phase with those identified here, and this flow field rotates about the centerline at a frequency consistent with the mean tangential velocity. It was identified that the central jet pushes the central recirculation zone down and stabilizes it, damping the precessing vortex core that is otherwise present without the central jet.

The residence times measured via scalar tracking are only valid under the condition that the sudden upstream impulse does not alter the downstream velocity field and the dominant flow features identified which contribute significantly to scalar transport. Therefore, the velocity time histories at multiple points in the flow have been inspected, and no significant step change in terms of both the mean and RMS velocities was found after the dye shutoff point. In addition, it is also important to check whether the step change alters the spectral content of the vortex pairs. An example of this check is shown in Fig. 7 for the no-jet and with-jet conditions, just at the exit of the annulus at a point through which the helical vortices pass. The resolution of the spectra changes in these plots depending upon the number of samples available before and after dye shutoff. For both cases, it can be seen that the dominant frequencies are not altered due to the transient in dye injection. Both with- and without-jet conditions are shown in Fig. 7 to highlight an important difference in the two flow fields, (other than the reduced upstream penetration of the central recirculation zone with the central jet). It is clear that the no-jet condition has a fundamental

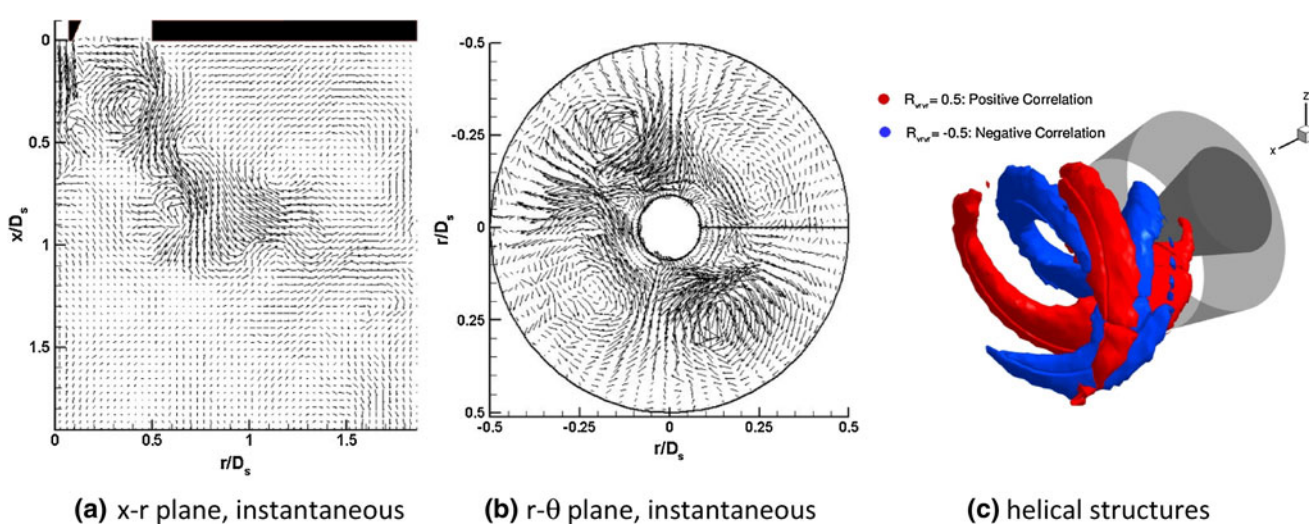


Fig. 6 Unsteady flow structures in fuel injector flowfields (Spencer et al. (2008))

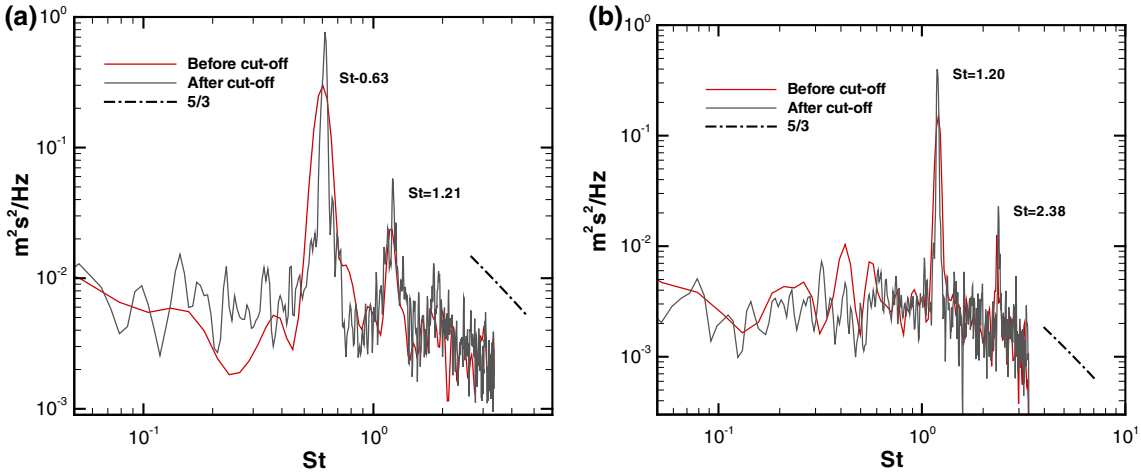


Fig. 7 PSD of axial velocity at $(x/D_s, r/D_s) = (0.04, 0.27)$

frequency close to a half of that with the central jet. This happens because the central jet stabilizes the flow field and makes it more symmetrical. This can be demonstrated using Fig. 8, which shows the spatial correlation fields of tangential velocity with respect to the velocity at the highlighted reference point which is located at the center of one of the strong vortices. In the no-jet case, there is little mirror symmetry of the 2 vortex pairs (about a vertical or horizontal line though the center point of Fig. 8a); thus, a periodic signal is only picked up once per revolution of this flow. With the central jet, symmetry of the left- and right-hand vortices and the top and bottom weaker vortices is evident, and a periodic signal is picked up twice per revolution. The possible interaction between these coherent structures and thermo-acoustic excitations makes this fuel injector an interesting test case from an instability point of view. Meier et al. (2007) present reacting data for the no-jet case at different fuel air ratios. It is found that at lower fuel injection rates, a self-generated combustion oscillation occurs at 290 Hz. Correcting for scale ($D_s = 27.8$ mm) and Reynolds number differences between experiments, this corresponds to a Strouhal number of 0.71 here, which is slightly higher than the isothermally deduced one of 0.63. This may be expected due to the acceleration effect that the heat release will have on velocity as the density reduces. At higher fuel injection rates, the flame stability improves because it moves away from the weak extinction but also possibly because the fuel injection jets are at a higher velocity, penetrate further into the swirl stream, and improved premixing occurs.

4.1 Time mean and RMS scalar fields

In Fig. 9, the mean velocity and RMS scalar fields are shown for the two cases. To indicate the locations of the

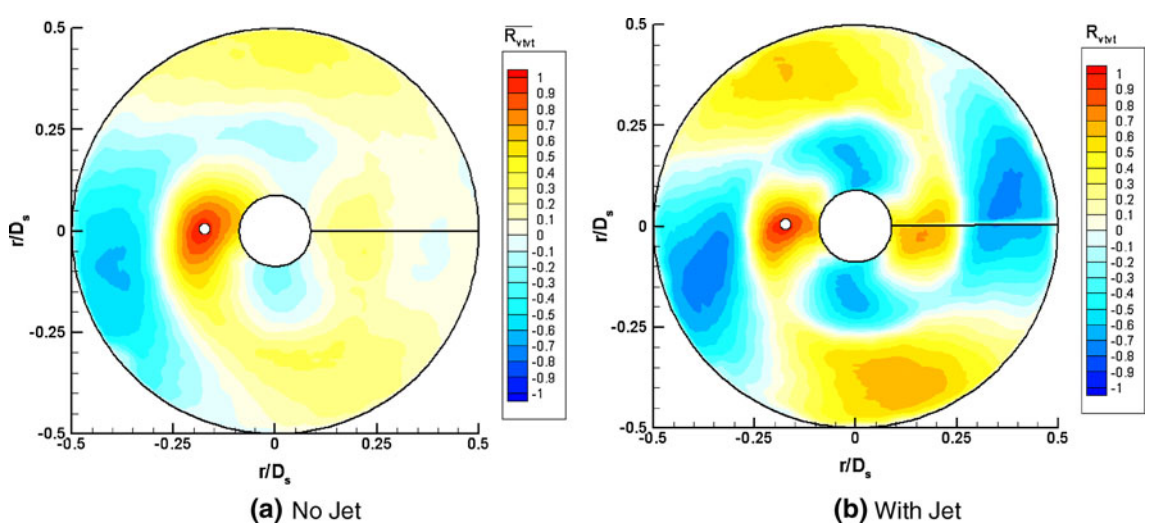


Fig. 8 Spatial correlation in tangential velocity at injector exit plane

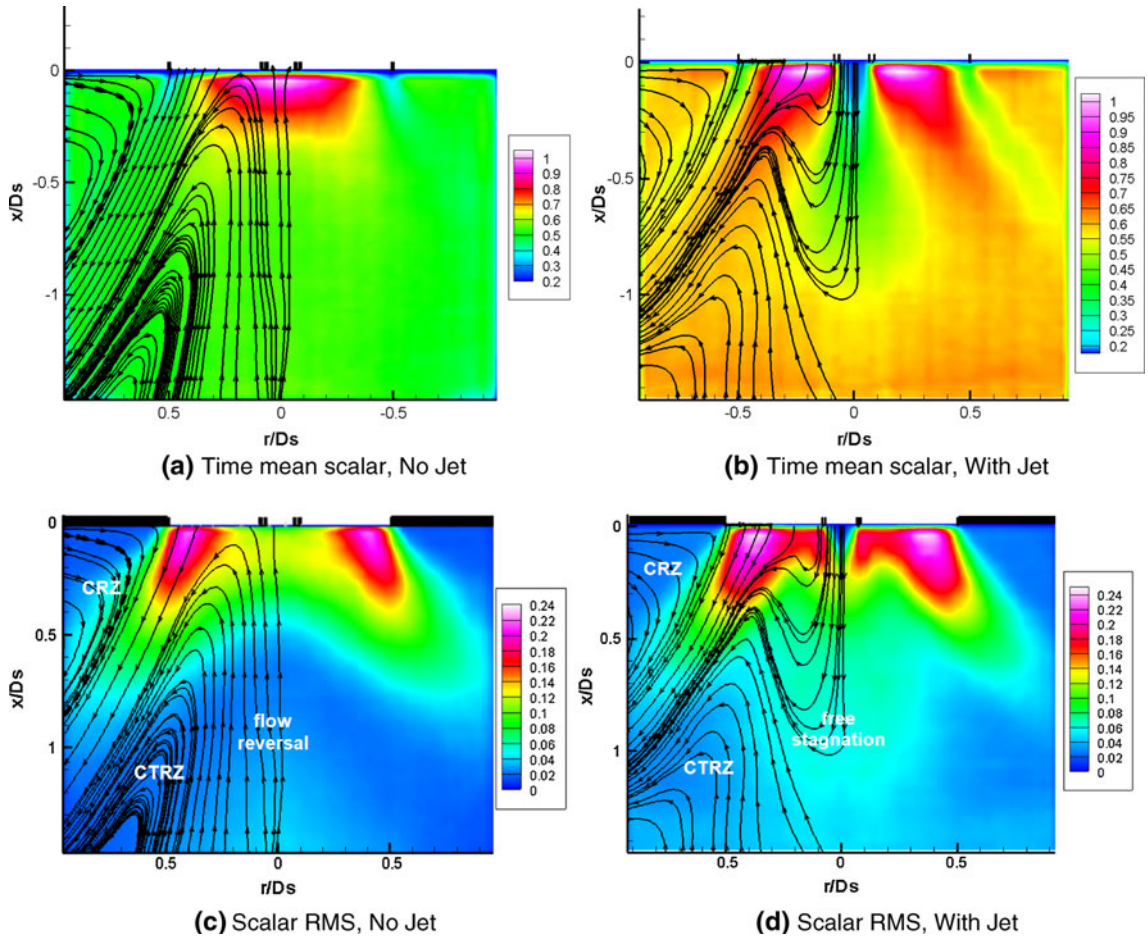
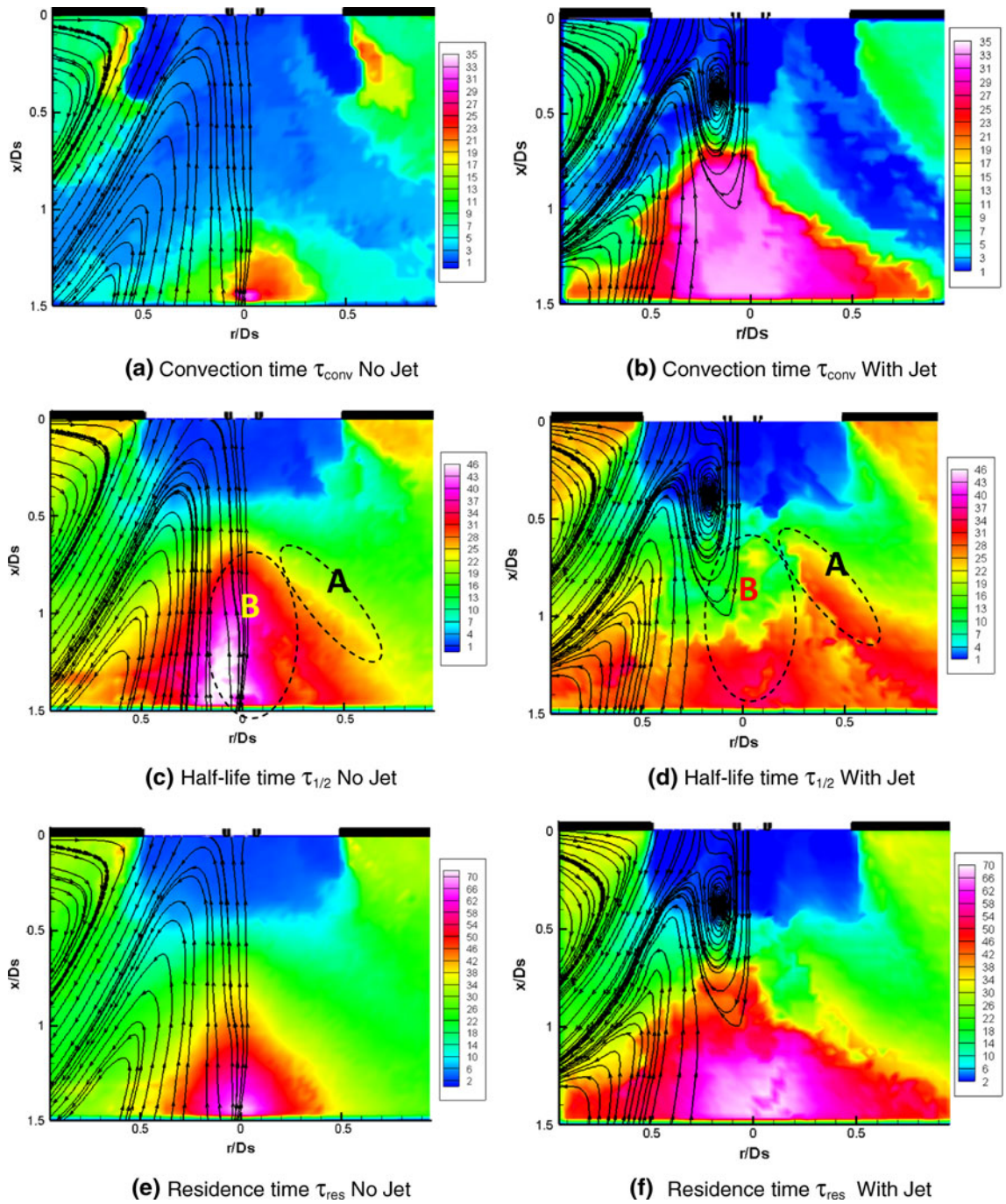


Fig. 9 Scalar statistics normalized by maximum value

swirl streams, as well as the central jet stream for the with-jet case, streamlines are shown on the left hand side of the mean scalar plots and RMS plots. The scalar has been non-dimensionalized in each case by the peak value in the FoV.

For the with-jet case shown in Fig. 9b, regions of high mean concentration are found just beneath the exit plane, bound by the inner shear layer ($r/D_s \sim 0.35$) of the swirl stream and the jet stream ($r/D_s \sim 0.07$) of undyed water. A slight backflow in the time mean is seen in this region, implying that localized high-temperature fuel-concentration fluid could be transported back into the swirl passage—a condition that could lead to flashback (Huang and Yang 2005). In addition, significant variation in scalar across the swirler exit also indicates incomplete mixing in the swirler passage, which is also seen in instantaneous images. The no-jet case as shown in Fig. 9a also shows high-concentration fluid bound by inner swirl cone and unmixedness at the swirler exit, though clearly the lack of the central jet does not wash the high scalar concentration away from the centerline, and the concentration is uniform in the radial direction below the central body. It may be noted that the high level of unmixedness at the swirler exit, which has also been observed in the CH_4 combustion data of Meier et al. (2007), is probably due to a short mixing distance from the premixing port to the swirler exit along with insufficiently penetrating fuel jets in the swirl slots as previously mentioned. The strong inner-radial bias of the fuel distribution is also seen in exit plane measurements ($r-\theta$ plane, taken at $x/D_s = 0.13$ not shown), but these demonstrated a good axis symmetry in the long time mean and were consistent with a cross-section at the same axial distance through all of the x–r plane measurements shown in Fig. 9.

High scalar fluctuations are found in the swirl stream shear layer for both cases as shown in Fig. 9c and d. These localized regions of high scalar fluctuation (larger than 20% of the mean) at the immediate swirler exit indicate the region where the helical vortex structures rapidly mix out the radial bias in scalar concentration such that the dye is more or less fully mixed at between 1 and 1.5 injector diameters downstream, where the scalar concentration values approach 0.5. In the with-jet case, the scalar RMS values are also increased along the centerline down to the



(a) Convection time τ_{conv} No Jet

(b) Convection time τ_{conv} With Jet

(c) Half-life time $\tau_{1/2}$ No Jet

(d) Half-life time $\tau_{1/2}$ With Jet

(e) Residence time τ_{res} No Jet

(f) Residence time τ_{res} With Jet

Fig. 10 Normalized characteristic times

free stagnation point. This is important because the magnitude of fluctuations compared to the local mean value is around 20–25%, whereas fluctuations in the no-jet case in the central recirculation zone are much smaller.

4.2 Characteristic residence times

The characteristic times evaluated for the two test cases are shown in Fig. 10. The convection time, τ_{conv}, for no-jet and with-jet are in Fig. 10a and b, respectively. For the with-jet case, the region along the centerline where the highest gradient of convection time is found ($x/D_s \sim 0.7$) is midway within the free stagnation region ($x/D_s \sim 0.5$ to 1.0) between forward flowing central jet and backward central recirculation zone (CRZ). The convection time of the with-jet case is markedly longer than that of the no-jet case because the central jet damps the PVC and stabilizes the CRZ, resulting in a steadier convective path, which

involves the swirl stream recirculating back into the FoV from downstream locations ($x/D_s > 1.5$). The convection path for the no-jet case, on the other hand, is subject to significant variation over time associated with a strong low-frequency PVC. The center of flow rotation orbits the geometrical centerline at a radius of around r/D_s of 0.1–0.2, and the swirling conical flow exiting the fuel injector is quickly entrained into the central recirculation zone. Hence, the no-jet case shorter convection time implies that disturbances that originate from inside the fuel injector are likely to be transported into the CRZ through large-scale turbulent features more quickly than the with-jet case. This difference may be exaggerated by the definition for τ_{conv}, which requires the scalar to fall by one standard deviation below the steady mean. It was seen in Fig. 9 that the with-jet case has higher scalar RMS values along the centerline than the no-jet case; thus, it will take longer at a given decay rate to exceed this condition. This is further discussed in the next section when looking at examples of decay curves. Supplementary videos are available of the scalar decay, which also demonstrates this behavior. The outer shear layer of the swirl stream (between it and the corner recirculation) also shows some differences between the two test cases, with the no-jet case having a longer convection time. This demonstrates that the corner recirculation regions have a different dynamic. This is explained because it is found that the reverse flow on the outer wall of the test section extends to x/D_s of 1.5 for the no-jet case, whereas the with-jet case corner recirculation is smaller, extending to x/D_s of 1.25 and is more stable in location.

The half-life times shown in Fig. 10c and d reveal two major differences, which are highlighted in the regions bounded by the dashed lines. First, the with-jet convection time in region (A) is up to 40% higher than the no-jet one. Region (A) falls within the swirl shear layer, featuring the vortex pairs, in which high scalar concentration fluid could be trapped. Indeed, the longer half-life time there in the with-jet case is consistent with the fact that the vortex pairs in this case are more coherent than those in the no-jet case as discussed earlier. The less coherent structures of the no-jet case dissipate and mix out more quickly and are entrained into the central recirculation zone, whereas for the with-jet case, the mixing between the coherent structures and the central recirculation zone is reduced. This is also consistent with the convection times discussed above, where scalar disturbances take longer to be entrained into the central recirculation zone. The second most striking difference is located in the region marked by (B), where the half-life times of the no-jet case are about 20–30% higher than the with-jet. This is believed to be due to the presence of a stronger PVC, as dye appears trapped in its low-pressure and velocity center, which precesses about the geometric centerline at a low-frequency as described above. This highest contour level on the centerline of Fig. 10c highlights the region over which the helical swirl center associated with a PVC is seen to exist through flow visualization. The central jet in the with-jet case helps to quickly dilute the scalar trapped in the top of the CRZ, but in both cases, the rest of the CRZ is relatively stagnant, and thus, the decay rates are relatively long.

Given the above differences in convective and half-life times, the residence times shown in Fig. 10e and f are found to be not too dissimilar. In general, the CRZ of the with-jet case is dominated by long convection times and short half-life times, while the no-jet case exhibits an opposite phenomenon. Both test conditions have essentially the same global timescale D_s/U_{ref} as defined in the introduction. An experimentally determined residence time distribution derived global timescale, τ_{res}, would therefore perhaps be expected to be similar. Although that happens to be the case here, the breakdown into a convective and decay component indicates a very dissimilar flow history. The combustion chamber sees inlet disturbances relatively quickly in the case of the no-jet case compared to the with-jet case, but then, the effect of the disturbance takes longer to decay from the chamber. It is interesting to note that the reacting no-jet case exhibits pulsating behavior under certain operating conditions as identified by Meier et al. (2007). This is not found to be the case when operating in the with-jet case, though both conditions have strong spikes in the flow spectra of near equal magnitude as seen in Fig. 7 (but no-jet frequencies are half of those in the with-jet case).

In the introduction, it is indicated that NOx production is dependent on residence time and also temperature. Thus, the difference in flow history—in particular how flow enters the central recirculation zone—could be quite important to the emissions in each case. In the with-jet case, the evidence suggests that the CRZ is predominantly filled from downstream which could be at reduced temperatures (due to wall cooling for example) compared to the near-field reaction zone. Thus, to fully understand emission production, an idea of the temperature field is also required. Some evidence of this is provided by Meier et al. (2007), but combining these measurement techniques for a reacting case is therefore potentially a valuable exercise to help determine where maximum production rates of emissions are in a given combustor geometry.

4.3 Examples of exponential fits

The single exponential function has been identified as the most suitable for describing the decay trend in swirl-stabilized flows and has been employed by a number of authors (Arai et al. 1998; Spencer and Cheng 2010; Read

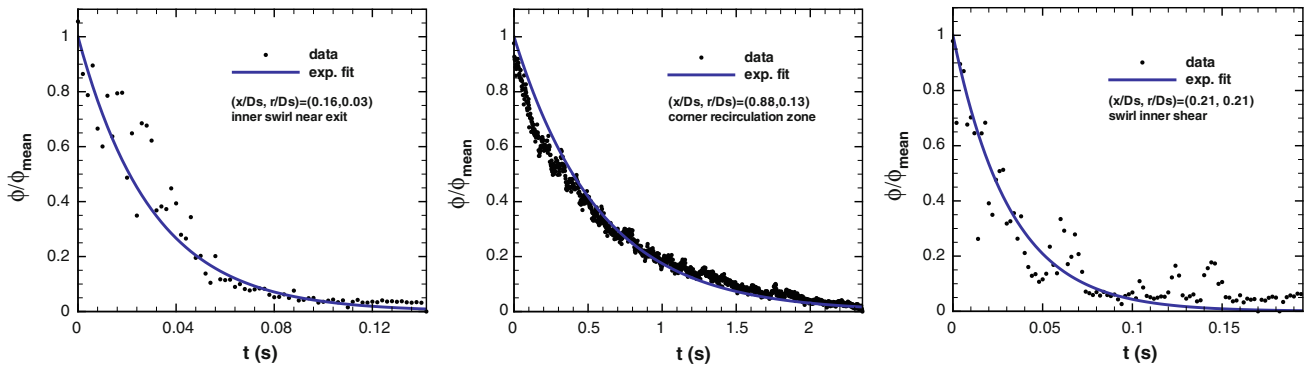


Fig. 11 Examples of reasonably good exponential fits to decay data (With-jet)

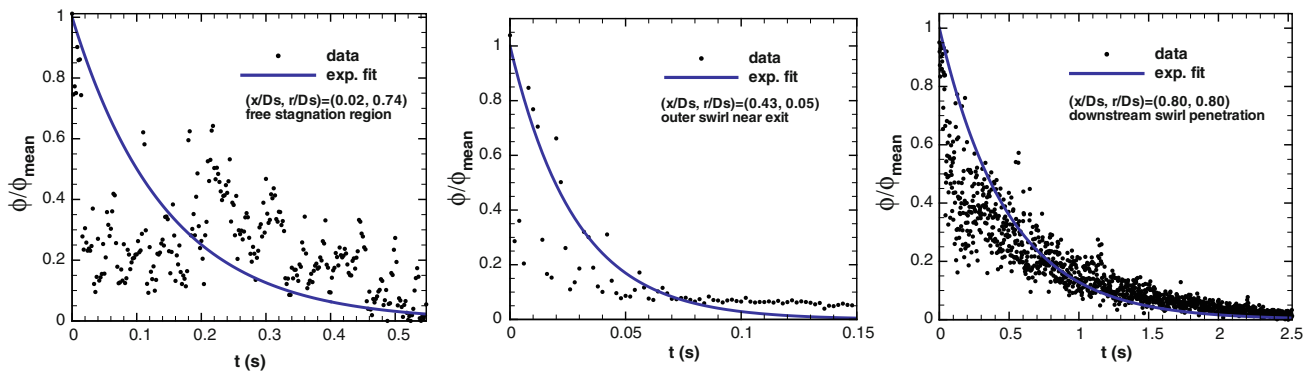


Fig. 12 Examples of poor exponential fits to decay data (With-jet)

and Hochgreb 2010). Examples of reasonably good exponential fits are shown in Fig. 11. Within the FoV explored for this flow configuration, the model decay function is a good estimate for around 60–70% of the region. However, due to strong localized turbulence, coherent vortex structures, and some unmixedness of the mixture at the swirler exit, the concentration decay can feature strong fluctuations in some regions which make the exponential best-fit seem crude. Figure 12 shows three examples of such crude fits. High scalar fluctuation in the free stagnation region is due to temporal variation of the jet penetration length; thus, the decay rate may be under predicted while the convection time over predicted. A crude fit is also seen at the swirler exit ($r/D_s = 0.43$), where the RMS scalar is at its peak (see Fig. 9). In these regions, bimodal distributions of the concentration have been found, and it can be seen that two separate exponential curves could better describe these data. This is consistent with the radial gradient in scalar at the swirler exit being started to be stirred by the emerging vortex pairs, which would cause intermittent high then low values of scalar. Further downstream in the swirl stream the bimodal distribution is mixed out, and the scatter in variation becomes more normal in nature.

4.4 Repeatability

Repeat measurements have been conducted for the no-jet case, and the timescales evaluated for these measurements are compared in Fig. 13 for a profile taken along the centerline. This direction has been chosen since it contains both the near field, dominated by high exit velocities (and therefore short timescales), and also the central recirculation zone, with low-velocity reverse flow (and associated long timescales). A good level of repeatability is seen, despite very high turbulence levels, particularly in the combined residence time. Discrepancy does occur in convection time and half-life. This appears to be due to a difference in the shape of the initial decay curve for the two measurements within the central recirculation zone. The low PVC precession frequency (St $\sim$ 0.014 using the definition in Sect. 2, see Spencer et al. (2008)) corresponds to a non-dimensional period of around 70 on these figures, so the phase angle of the precession is probably becoming important to the measurement and affecting where the algorithm determines the intercept point that defines the start of the decay. This region is where the largest discrepancies have occurred in all measurements reported

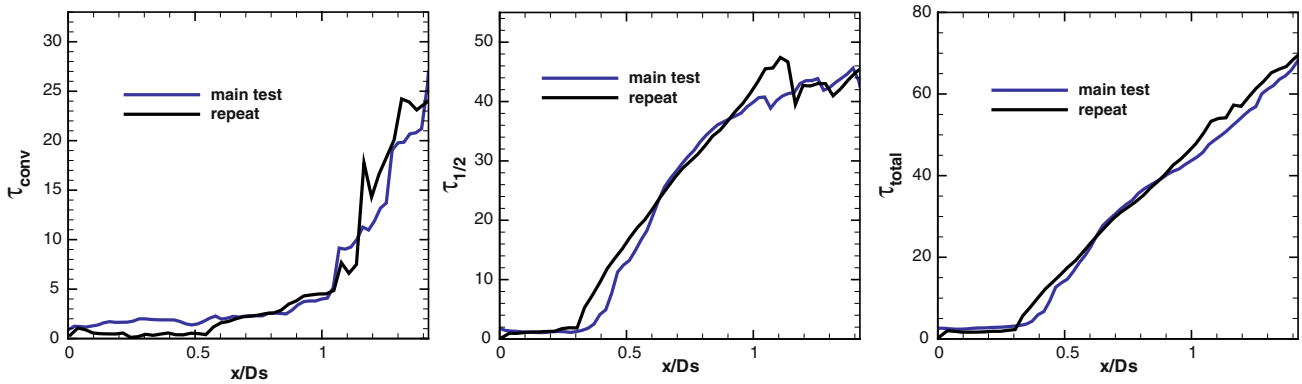


Fig. 13 Repeatability: Characteristic times along centerline (No-jet)

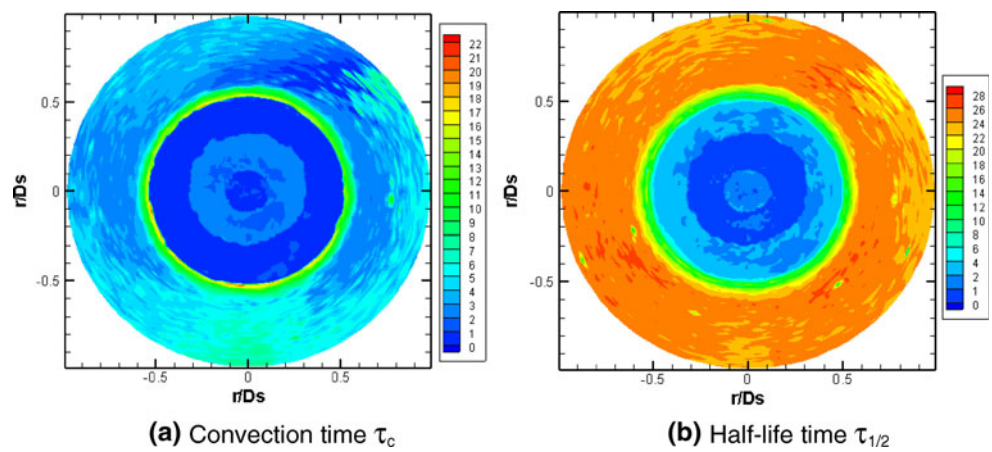


Fig. 14 Measurements of residence times in the r–θ plane at $x/D_s = 0.13$ (with-jet case)

here and demonstrates that with a low number of parameters describing the decay (τ_{conv}, $\tau_{1/2}$) single measurements are probably acceptable providing the model decay curve is representative and that the derived parameters are longer or at least the same order of magnitude when compared to the flow timescales being investigated.

The level of circumferential variation is also a good test of repeatability in the sense that over a long time average, the velocity and scalar fields are found to be axis symmetric; thus, it would also be expected of the scalar timescales. Looking close to the exit plane of the injector ($x/D_s = 0.13$) on an r–θ plane is now a rotating vortex pattern that has a non-dimensional rotation period of about 1.6 (see Fig. 6b). In general, it can be seen from Fig. 14 that the circumferential variation in timescales does not exceed ±5% in the swirl stream exit ($r/D_s < 0.5$). In the corner recirculation region for $r/D_s > 0.5$, the variation is typically less than 10%, but here, the flow timescales are an order of magnitude larger than for the swirl stream exit.

The data in Fig. 14 have been circumferentially averaged to produce the radial profile of characteristic times in Fig. 15. This process allows for the best estimate of the average of multiple measurements in order to remove variations due to the flow structure rotational phase angle. The near-zero convection times in Fig. 15a and c indicates the region around the time reference point at the swirl stream exit has uniform values and is a reliable base time for the time at which the dye shutoff enters the combustion chamber. The circumferentially averaged profiles are compared to an equivalent profile extracted from the x–r plane measurement at $x/D_s = 0.13$ (which can contain no circumferential averaging). There is good agreement between the two with-jet tests on perpendicular planes, demonstrating good repeatability of the technique. The agreement in the no-jet case is also reasonable, but the very long timescale PVC that exists in this condition has reduced the level of agreement that is seen in the with-jet case. This could of course be improved by performing multiple realizations if the additional accuracy is required.

5 Conclusion

This paper describes an experimental technique using combined PIV/PLIF that has been used here to quantify residence time distribution in a swirling flow representative

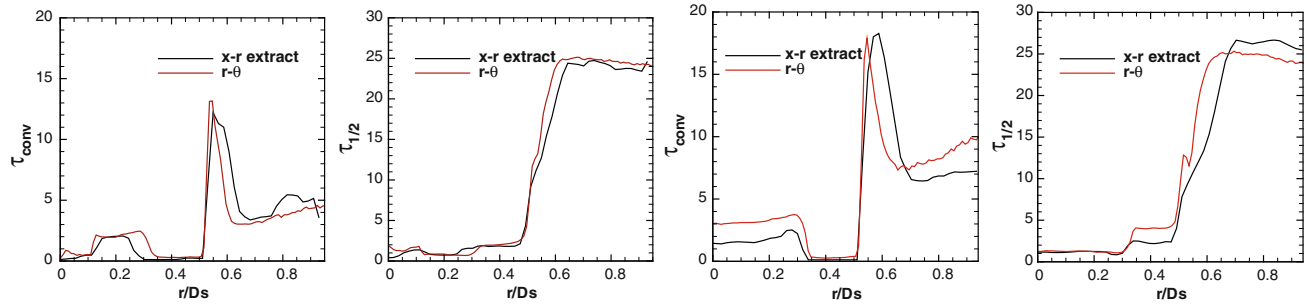


Fig. 15 Convection and half-life times on x–r and r–θ plane measurements at $x/D_s = 0.13$

of that found in partially premixed combustors. The technique has only been performed in isothermal flows in water, but it could be adapted for reacting flows in air in the future. It relies on looking at the temporal response to the cessation of the injection of a conserved scalar at the inlet to a particular region of interest. This has been described in literature previously for point-based measurements, but here, planar measurements are performed providing good spatial resolution of timescales within a region of interest.

The measured decay curves are characterized using a model decay function consisting of a convection time and half-life time: The former describes the shortest time span by which a disturbance at the fuel injector exit can be transported to the point of interest, and the latter describes the rate of exponential decay after the reduction in concentration due to fuel shut is transported to the point of interest. The residence time (time to reach half of the initial value) can be considered to be the sum of these two quantities. Residence timescales are important in determining NO_x emissions, for example, and many correlations exist which relate emission index to residence time and temperature. With a spatially resolved statistical description of the residence time, along with some knowledge of the temperature distribution, it may be possible to identify the main regions of emission production. The technique has been applied to the study of the flow field downstream of a radially fed swirl fuel injector in the vortex breakdown region, and a comparison was provided between the with-jet and no-jet cases.

It was found that for the no-jet case, fluid is likely to be transported to the center recirculation zone ($x/D_s \sim 0.5$ to 1.2) much more quickly than the with-jet case due to the former's relatively unstable nature. The with-jet case contains two pairs of very coherent helical vortices along with a more stable central recirculation zone, and the rate of mixing between these two regions is seen to be reduced. Thus, in the with-jet case, the CRZ is filled more from the downstream direction rather than direct turbulent transport through the shear layer between it and the swirl cone that surrounds it. In the same region, it was also found that the half-life times of the no-jet case are significantly higher, associated with the presence of the PVC that appears to trap fluid in the swirl center.

Once inlet disturbances had been convected to a location of interest, it was found that an exponential decay function was a good model to the actual decay for around two-thirds of the locations investigated. Demonstrating the technique in a high-swirl combustor flow is the motivation of this work, but there are wider potential combustion-related applications of the method. Bluff body flame stabilization for example requires the determination of wake characteristic timescales. This has previously been demonstrated by filling wakes with smoke and using whole-field, non-calibrated light attenuation techniques (Winterfield 1965)—a measurement which could be significantly improved using the techniques described here.

References

Adrian RJ (1991) Particle-imaging techniques for experimental fluid mechanics. Ann Rev Fluid Mech 23:261–304

Arai M, Nomura T, Hiroyasu H (1998) Residence time measurement in a swirl type combustor with helium tracer. Jap Soc Mech Eng 86(1302A):2619–2625

Cheng L, Dianat M, Spencer A, McGuirk JJ (2010) SPIV/PLIF validation of LES predictions of scalar mixing in high-swirl fuel injector flows. In: ETMM8: 8th Int. ERCOFTAC Symp. Marseille, France. (Invited to Jnl Flow Turb. And Comb.)

Chigier NA, Beer JM (1972) Combustion aerodynamics. Applied Science, England

Di Nardo A, Calchetti G, Mongiello C, Giammartini S, Rufoloni M (2009) CFD Modelling of an experimental scaled model of a trapped vortex combustor. In: Proceedings of the European combustion meeting

Driscoll J, Chen RH, Yoon Y (1992) Nitric oxide levels of turbulent jet diffusion flames: effects of residence time and Damkohler number. Comb Flame 88:37–49

Dunham D, Spencer A, McGuirk JJ, Dianat M (2009) Comparison of unsteady reynolds averaged navier–stokes and large eddy simulation computational fluid dynamics methodologies for air swirl fuel injectors. J Eng Gas Turbines Power 131:011502. doi: 10.1115/1.2969096

Giordano D, Giammartini S, Manfredi F, Guj G (2001) Non dimensional numbers dependency on a Dry Low NOx Annular

Burner Analysed with PIV technique. In: 9th National Conference AIVELA, Ancona, 2001
Heywood JB, Fay JA, Lindern LH (1970) Jet aircraft air pollutant production and dispersion. AIAA 70-115
Huang Y, Yang V (2005) Effect of swirl on combustion dynamics in a lean-premixed swirl-stabilized combustor. Proc Comb Inst 30:1775–1782
Janus B, Dreizler A, Janicka J (2007) Experiments on swirl stabilized non-premixed natural gas flames in a model gasturbine combustor. Proc Comb Inst 31:3091–3098
Lieuwen T (2006) The gas turbine handbook. Department of Energy, USA
Meier W, Weigand P, Duan XR, Giezendanner-Thoben R (2007) Detailed characterization of the dynamics of thermoacoustic pulsations in a lean premixed swirl flame. Comb Flame 150
Midgley KK (2005) An isothermal experimental study of the unsteady fluid mechanics of gas turbine fuel injector flowfields. Ph.D. thesis, Loughborough University, UK
Midgley K, Spencer A (2005) Unsteady flow structures in radial swirler fed fuel injectors. ASME J Eng Gas Turbine Power 127:755–764
Noble D, Zhang Q, Shareef A, Tootle J, Meyers A, Lieuwen T (2006) Syngas mixture composition effects upon flashback and blowout. ASME 2006-90470
Ozdemir IB, Peters N (2001) Characteristics of the reaction zone in a combustor operating at mild combustion. Exp Fluids 30:683–695
Pantano C, Sarkar S, Williams FA (2003) Mixing of a conserved scalar in a turbulent reacting shear layer. J Fluid Mech 481:291–328
Read R, Hochgreb S (2010) Isothermal flow measurements in a gas turbine combustor using a fast flame ionization detector. Meas Sci Technol 21:1–10
Roffe GK, Venkataramani KS (1978) Emissions measurements for lean premixed propane/air systems at pressures up to 30 atmospheres. NASA CR-159421
Spencer A (1998) Gas turbine combustor port flows. Ph.D. thesis, Loughborough University, UK
Spencer A, Cheng L (2010) Combustor residence time measurement using planar laser induced fluorescence. In: Proc ASME Turbo Expo GT2010-22573
Spencer A, McGuirk J, Midgley K (2008) Vortex breakdown in swirling fuel injector flows. ASME J Eng Gas Turbines Power 130:021503.1–021503.8
van der Lans RP, Glarborg P, Dam-Johansen K, Larsen PS (1997) Residence time distributions in a cold, confined swirl flow: implications for chemical engineering combustion modeling. Chem Eng Sci 52:2743–2756
Winterfield G (1965) On processes of turbulent exchange behind flame holders, The Combustion Institute, pp 1265–1275